SHINING 3D

RobotScan Control

V2.1.1

User Manual

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Overview

Welcome

This user manual (hereinafter referred to as "the Manual") introduces the operation of RobotScan Control.

Symbol Conventions

Symbol	Meaning
	Note: This symbol is used to inform you of the additional information of the product.
	Caution: This symbol is used to inform you of incorrect operations that may damage the device or result in data loss. Any damages resulting from misuse are not covered by the warranty.
	Warning: This symbol is used to inform you of the potential risks that may result in serious personal injury and other safety incidents.

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Quick Guide

This chapter provides an overview of the RobotScan Control software, helping you quickly find the relevant operational assistance content.

Here you can learn about the software's functions, installation, and other related content.

→ What kind of software is this?

→ How to install the software and what are the environmental requirements?

→ General introduction to the software interface

Once you have successfully installed the software, the complete process for using the software is as follows:

Add Device

Arrange Workflow

Before designing the automation workflow, you need to add and connect the device.

→ How to add and connect the device?

After connecting the device, design the workflow and execute the automation tasks.

→ How to arrange the automation workflow?

Hardware

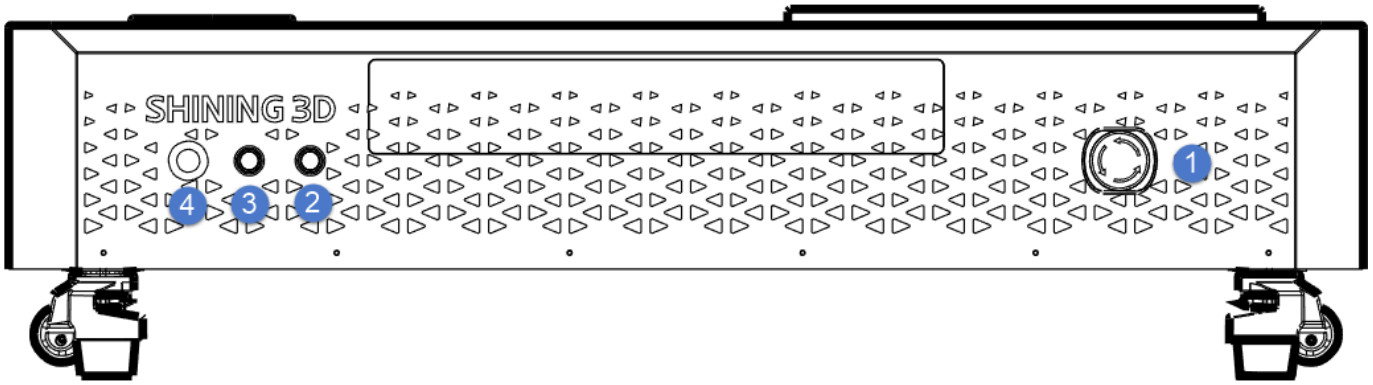
Device Appearance

Caution

The fixture and workpiece are not included in the parts list. Please provide them yourself.

CobotBase

Front



① Emergency Stop Button

Pressing the **Emergency Stop** button will end the process, and the currently executing command will be displayed in red. The status bar at the bottom of the software will show the current status in real-time.

After the emergency stop, if no recovery message is received, the **Start**, **Pause**, and **End** buttons on the software programming interface will be disabled. Once the recovery message is received, the **Start** button on the interface will be enabled.

② Start Button

Scenario	Action	Function Description
When connected to software	**Short press**	Equivalent to the **Start** operation in the software programming interface. The robot will start executing from the currently selected command.
When not connected to software	**Long press for 5 seconds**	The calibration board will automatically pop out or retract. This function only responds when the CobotBase is not connected to the software. When the CobotBase is connected to the software, this action will have no response.

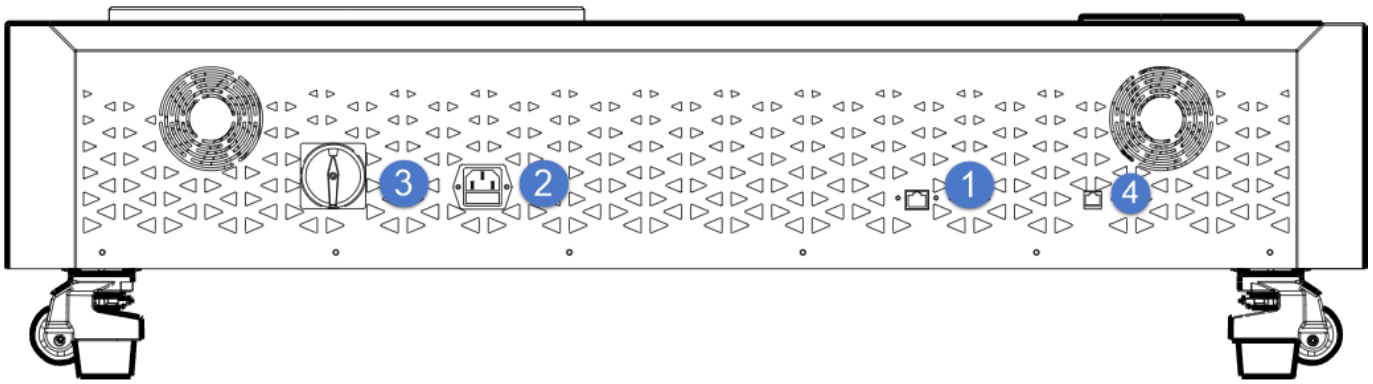
③ Reset Button

Scenario	Action	Function Description
When connected to software	**Short press**	The software will pop up a confirmation window asking if you want to reset the robotic arm, turntable, and clear the scan data.
When connected to software	**Long press for 5 seconds**	Immediately performs the following actions: resets the robot, resets the turntable, and clears all scan data. • **During process execution:** If there is an ongoing task or process, long pressing this button will immediately terminate the current execution and perform the aforementioned reset and data clearing operations. • **When no process is executing:** If there is no ongoing task, long pressing this button will directly execute the reset operation. If there was previous scan data, it will also be cleared.
When not connected to software	**Long press for 5 seconds**	The robot will reset to the preset position. This function only responds when the CobotBase is not connected to the software. When the CobotBase is connected to the software, this action will have no response, and the software will take over the function of the reset button.

④ Power Indicator Light

After powering on the CobotBase, the power indicator light will remain on.

Back



① Ethernet/Network Interface

Used to connect the network cable for wired communication between the CobotBase and software, robot, and other devices. Compatible with `OptimScan Q12 or Q9`

② Power Input Interface

Input: 100~120V / 200~240V; 50/60 Hz; 370W Max

Caution

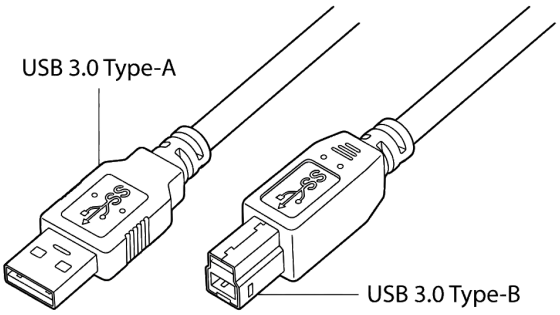
- Please select the power input that matches your local voltage.
- Please ensure to use a power cable that meets the voltage and current standards, and plug it in securely.

③ Power Switch

Used to control the overall power on and off of the CobotBase.

④ USB 3.0 Type-B Interface

Connect USB 3.0 Type-B to the computer's USB 3.0 Type-A interface.



Software

Introduction

RobotScan Control is an automated software platform that seamlessly integrates 3D scanners, collaborative robots. Its visual workflow design let users build automation processes without programming. It can continuously captures 3D data, connects with inspection software, and automatically generates standardized reports — simplifying and streamlining high-precision 3D scanning and batch inspection.

Software Applications

Aerospace

Automotive Industry

Inspect the shape and dimensions of high-precision components and complex surfaces to ensure product safety and reliability.

Conduct regular 3D inspections during the production process to monitor the stability of the production line and the consistency of product quality.

3C Electronics

Convert actual parts into digital models through 3D scanning and measurement for product design modifications, fault analysis, and more.

Precision Hardware

Perform full-size inspections and tolerance analysis on precision machined parts to ensure machining accuracy and product qualification rates.

Precision Injection Molding

Achieve size inspection and deformation analysis of molds and injection parts, aiding in product quality improvement and process optimization.

Software Installation

Operating Environment

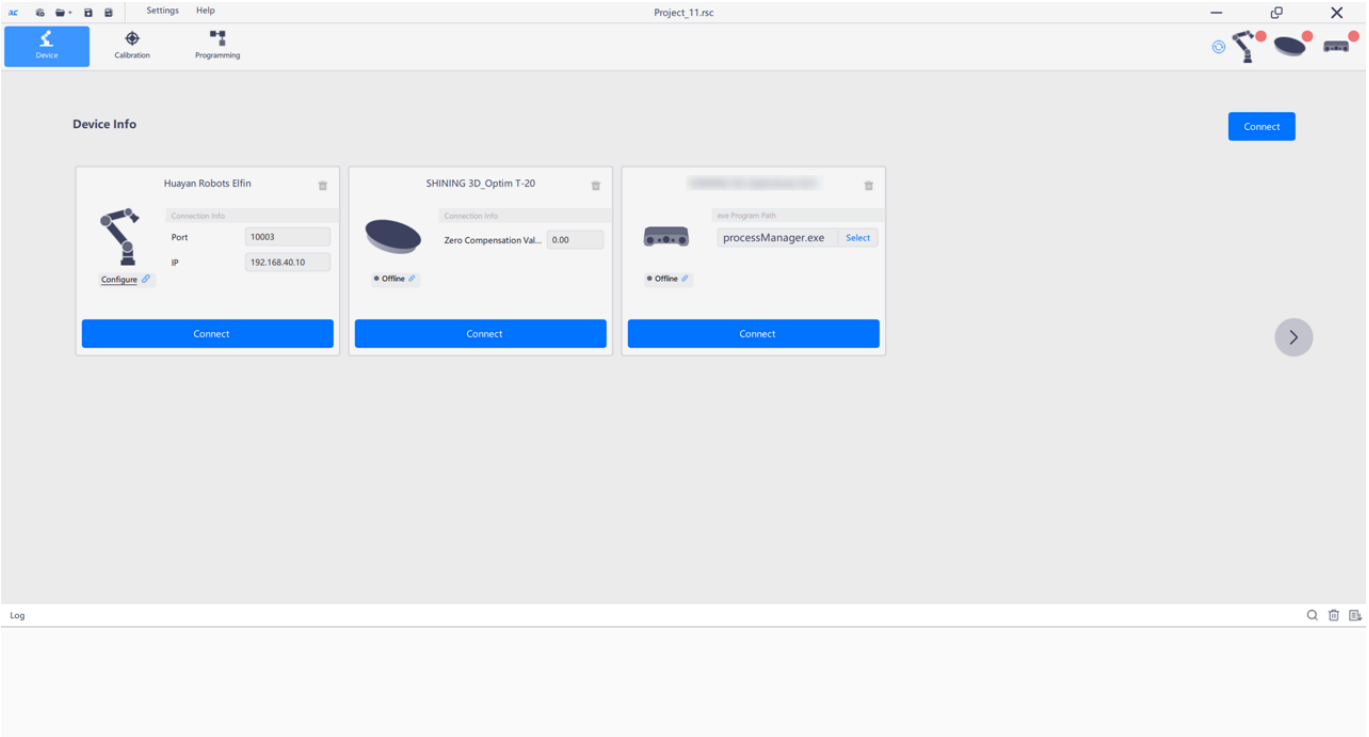
Recommended Configuration	
Processor	Intel® Core™ i7-13700H or higher
Graphics Card	NVIDIA RTX 4060 Laptop GPU or higher
Video Memory	8 GB or more
RAM	64 GB or more
Operating System	Windows 11 (64-bit)

Note

- To ensure the software runs properly, please use an NVIDIA graphics card.
- Administrator privileges are required for software installation. The initial installation may take some time, so please be patient.
- Do not install the software in the **C:\Program Files or C:\Program Files (x86)** directories to avoid issues with permissions that may prevent the software from starting properly.

Software Interface

Overview



Menu Bar

① File

Supports **New**, **Open**, **Save**, and **Save As Solution** (*.rsc).

To open recent files, click on to the right of , which will expand the recent files list.

Note

- The default name is **Project_1**, and each time a new solution is created, the number increments by one.
- The default path is **C:\Users\Current Logged-in User's Username\Documents\RobotScan Control\Solution**, which can be modified in **Settings**.

② Settings

Setting Item	Description
Select Language	Change the interface language, supporting Chinese, English, and German.
Project Save Path	Modify the default path for the project. Folders requiring administrator permissions are not supported.
Automatically start at boot time	/
Import last program from previous project when creating new project	/

③ Help

User Manual: Click to open this user manual.

Shortcut operations: For details, see Shortcut Operations.

Log

Display execution information, formatted as `yyyy-MM-dd HH:mm:ss log level log message`.

Device Configuration

Used to add, configure, and connect devices; after connecting the device, you can view the device connection status.

You can save the current solution as a template for quick application.

Calibration

To calibrate the device's scanning accuracy and maintain the quality of the scans.

Programming

To create and manage workflows, utilize the visual programming interface. Drag-and-drop nodes onto the canvas and connect them to design complex scanning and inspection routines.

Shortcuts

Global	
Save Project	^ Ctrl + S
Copy	^ Ctrl + C
Paste	^ Ctrl + V
Cut	^ Ctrl + X
Delete	⌦ Del
Group	^ Ctrl + G
Ungroup	^ Ctrl + ⌫ Backspace
Rename Group	F2
Node View & Scan View	
Zoom	Scroll Wheel
Pan	Middle Mouse Button
Vertical Scroll	^ Ctrl + Scroll Wheel
Horizontal Scroll	⇧ Shift + Scroll Wheel
Multi-select	⇧ Shift + Left Click
Box Select	Left Click Drag
Inspection Report	
Zoom	^ Ctrl + Scroll Wheel

Device Management

Device Management

RobotScan Control supports the connection and control of various models of robots, scanners, turntables, and other devices.

Add Device

After creating a new project, a scanner, turntable, and robot are automatically added by default.

Connect Device

1. Configure device information.

   - **Huayan Robot**: Enter the IP address 192.168.40.10 (the default port is 10003 and cannot be modified).

     > **Note**
     >
     > - The computer's network port must be in the same subnet as the robot's IP address.
     > - After entering the IP address, click **Configure** to open the robot's backend management page (the entered IP address + dist, such as 192.168.40.10/dist).

   - **Turntable**: Enter the zero offset (i.e., the offset value between the target angle and the real-time angle).

     > **Caution**
     >
     > Modifying the zero offset will affect subsequent turntable programming.

   - **Scanner**: Select the local path for the scanning software.

2. Click the **One-Click Connect** button, and after the prompt **Verification Successful**, the device will connect automatically. If the verification fails, please try again.

Icon	Status
	Not Connected
	Connection Successful

Note

- Before connecting the scanner, please ensure that the device has been authorized and activated.
- If the scanner is offline, the software will automatically attempt to reconnect.

Caution

After the robot is successfully connected, it is not enabled. You need to click **Start**, and after waiting for the robot to power up completely, continue by clicking **Enable** to use it.

Text	Status	Description
Open	Successfully connected to the robot, the robot is powered off	Click to power on the robot
Enable	Successfully connected to the robot, power on the robot successfully	After clicking, the robot can perform teaching and execute workflows
Enable (Disable)	Robot connected successfully, robot is enabled	After clicking, the robot cannot move
Close	Successfully connected to the robot, power on the robot successfully	Click to turn off the robot power

3D Mouse

You can use the mouse to control the robot for quick translation and rotation operations during robot point recording, replacing manual teaching.

For more information, please refer to the 3Dconnexion User Manual .

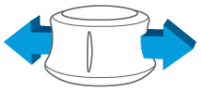
Mouse Connection

Plug the USB receiver into the USB port of your computer, then you can use the SpaceMouse Wireless in the software.

Operation Instructions

Caution

- The 3D mouse can only control the robot during point recording.
- Left-click to save the point position, right-click to exit recording.

Illustration	Description
	Control by pushing the button left and right: • Robot end, X positive and negative translation • Single-axis movement, first joint axis
	Control by pushing the button forward and backward: • Robot end, Y positive and negative translation • Single-axis movement, second joint axis
	Control by pushing the button up and down: • Robot end, Z positive and negative translation • Single-axis movement, third joint axis
	Control by tilting the button left or right: • Robot end, RX positive and negative translation • Single-axis movement, fourth joint axis
	Control by tilting the button forward or backward: • Robot end Y-axis rotation, RY positive and negative translation • Single-axis movement, fifth joint axis
	Control by rotating the button: • Robot end X-axis rotation, RZ positive and negative translation • Single-axis movement, sixth joint axis

Calibration

Calibration Instructions

You can re-adjust parameters of the scanner through calibration, which ensures not only the accuracy of the device, but also the scanning quality.

Note

Calibration is required under the following conditions:

- The scanner was severely shaken or vibrated during transportation.
- The scanner is used for the first time or it is not in use for a long period of time (1-2 weeks).
- The accuracy drops during scanning, such as frequent alignment failures or the inability to recognize markers.
- Incomplete data is acquired or the quality of scanned data drops during scanning. severely decreased, such as frequent stitching errors or failure to recognize markers.

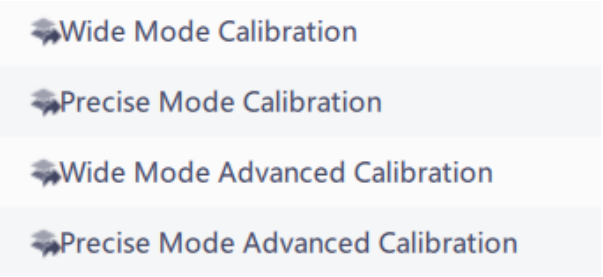
Caution

- Do not wipe the calibration board with chemical liquids.
- Do not put heavy objects or sundries on the calibration board.
- After using the calibration board, put it in the flannel bag.
- The calibration board can only be used for the calibration of the device.
- Keep the calibration board away from corrosives, metals and sharp objects to avoid corrosion or damage.
- Make sure that the markers on the calibration board aren't damaged or stained, and that the front side of the calibration board is clean and free of scratches.
- Make sure both the scanner and tracker are online before or during calibration.
- After powering on the device, please heat it to a proper temperature (about 35℃) before the calibration to ensure accuracy and precision.

Calibration Operation

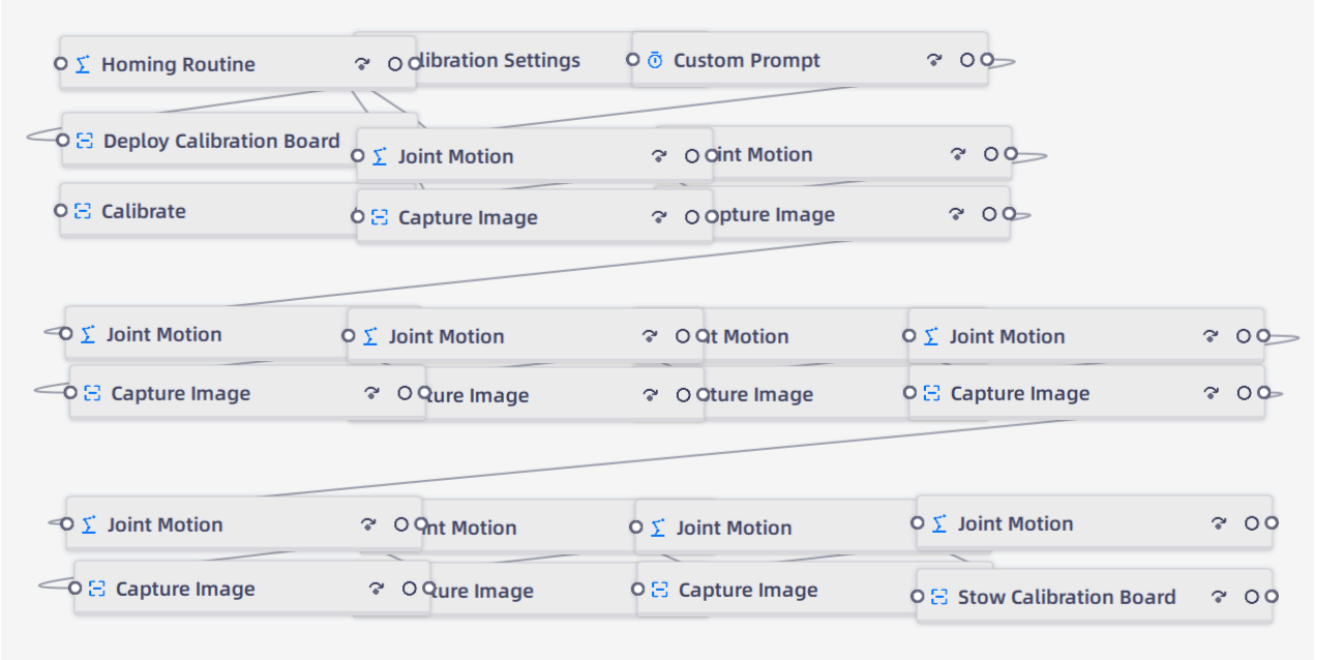
Please follow the steps in the interface calibration guide.

Calibration includes four types: **Wide Mode Calibration**, **Precise Mode Calibration**, **Wide Mode Advanced Calibration**, and **Precise Mode Advanced Calibration**. Each type has a corresponding calibration template.



Taking **Wide Mode Calibration** as an example:

1. Drag the calibration template onto the canvas to automatically generate the calibration command.



> ⚠ **Caution**
>
> - Modifying the parameters of commands from the template will not be saved to the template; you need to manually save the solution.
> - Modifying the parameters (calibration mode/calibration range) of the Calibrate command created from the template will not regenerate the corresponding commands; you need to delete the current commands and reselect the desired template.

2. Click ▶ to start executing the command.

> ⚠ **Caution**
>
> Calibration for different range requires different calibration boards; replace the calibration board manually when changing the range.

3. Calibration is complete once the prompt indicates successful calibration.

> **Note**
>
> - If the calibration is unsuccessful, you can modify the **joint movement** command's points and try again.
> - If calibration still fails after multiple attempts, please contact the supplier or technical support.

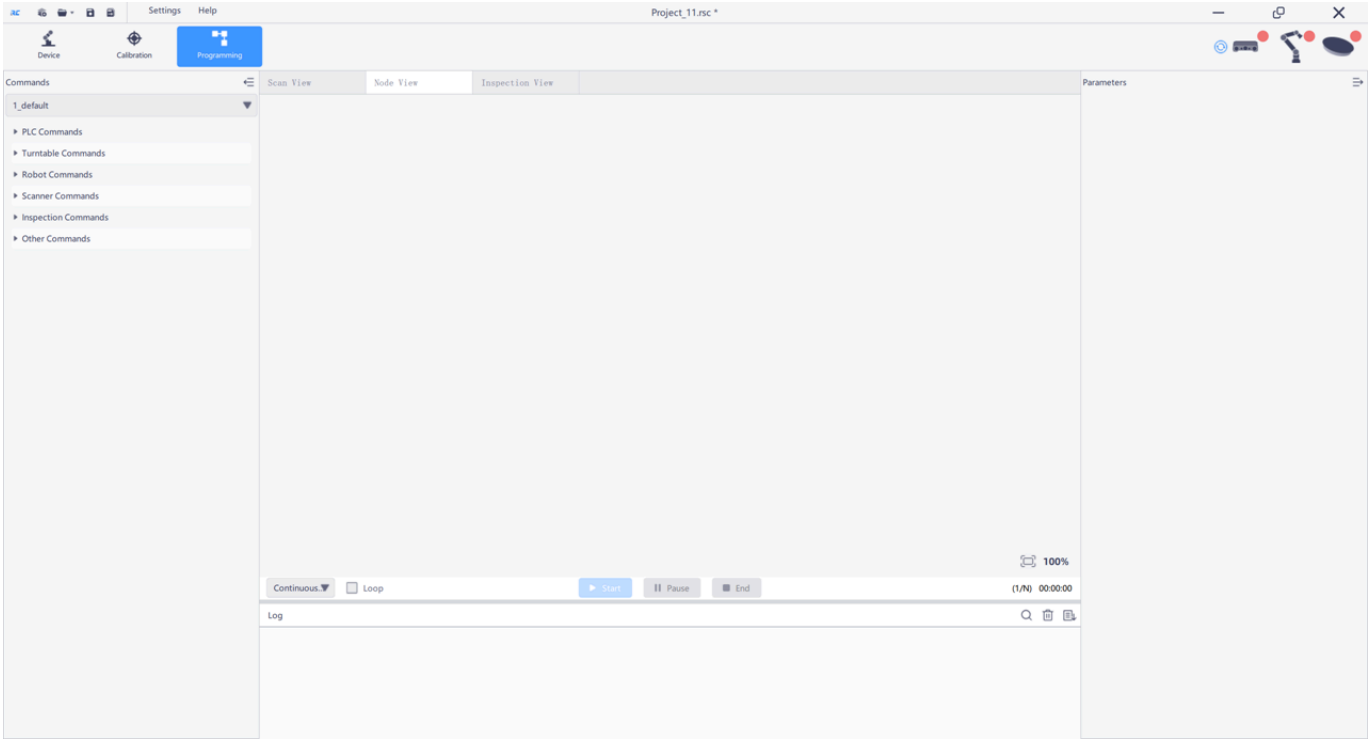
Programming

Streamline your scanning and inspection tasks with Robotscan Control's powerful workflow engine. Visually design, connect, and edit commands with simple drag-and-drop actions.

Example

Rapid Scan Template

Interface



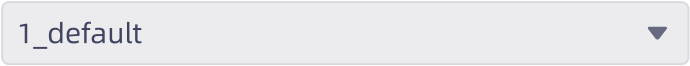
Workspace

View: A view is a tab, and you can drag the view to become a separate window.

- Node view: Used to place and connect commands.
- Inspection View: Used to view the detection report.
- Scan view: Used to view scan results.

Command List: A list of all available commands from which you can drag commands onto the canvas.

Program List: Used to save and manage the commands you have assembled. You can view, create, or import different program files for easy reuse and sharing.



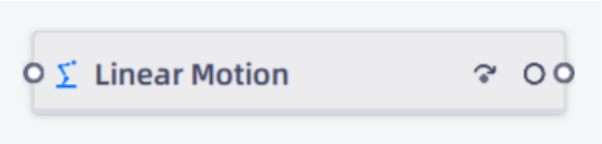
Parameter Panel: Displays the parameter settings of the selected command.

Command Interface

All commands in Robotscan Control are based on a similar structure. This interface is suitable for any type of command.

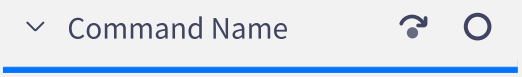
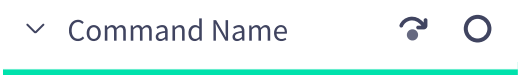
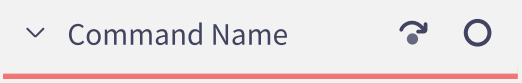
Title

The title displays the name/type of the command. On the right side of the title are the skip and breakpoint buttons for debugging.



Status

Below the title, the current status of the command is displayed, including not executed, executing, execution completed, and execution failed.

Status	Example
Not Executed	
Executing	
Execution Completed	
Execution Failed	

Input

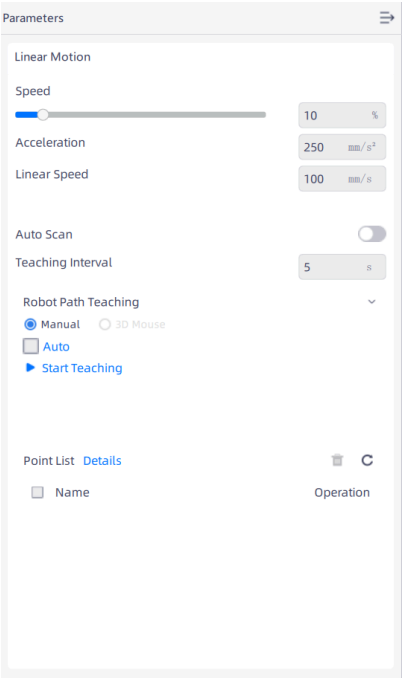
The input is on the left side of the command.

Output

The output is on the right side of the command and can be connected to the input of another command.

Parameters

Many command parameters may affect how they interact with input and output.



Parameter for the robot's joint motion command.

Operations

Create Command

Drag the command you need from the command list onto the canvas.

Connect Commands

There are two ways to connect commands:

- Select the output port of one command on the canvas, drag it to the input port of another command, and release the mouse to complete the connection.
- Click on a port to create a connection line, move it to a blank area on the canvas, click again to bring up the command list, and select a command to complete the connection.

Some commands can accept multiple inputs or outputs; you can connect multiple commands to the same command or connect the output of one command to multiple commands.

Edit Command

Click on a command on the canvas, or use `⇧ Shift` + `left-click` / drag to select multiple commands, then right-click to choose an action from the menu: step execution/grouping/copying/pasting/cutting/deleting, or use the shortcut keys.

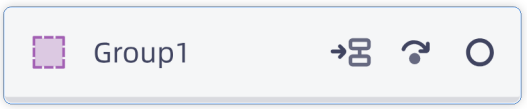
Name	Shortcut Key
Step-by-Step Execution	/
Group	`^ Ctrl` + `G`
Copy	`^ Ctrl` + `C`
Paste	`^ Ctrl` + `V`
Cut	`^ Ctrl` + `X`
Delete	`⌦ Del`

> **Note**
>
> Connections between commands will be copied along with the commands.

Select a command on the canvas and edit the command parameters in the parameter panel on the right.

Group Command

To simplify complex workflows and achieve repetitive tasks, Robotscan Control allows users to group multiple commands.



Grouping operation: Select multiple commands on the canvas (supports multi-selection or box selection), and complete the grouping by choosing **Group** from the right-click menu or using the shortcut `^ Ctrl` + `G`.

Loop Control: After grouping commands, you can set the number of times the commands within the group will be executed through the parameter panel on the right.

Renaming a group: After selecting the group, you can right-click or press the F2 shortcut key to rename it.

Edit Group: Click the edit group command icon to view and edit the commands within the group in a new view.

Execute Command

1. Choose the execution method:
   - **Step-by-Step Execution**: Execute the currently selected command and stop. This mode is typically used for debugging purposes, allowing you to execute each command in the workflow step by step, checking the output at each step to identify errors or optimize the process.
   - **Continuous Execution**: Execute the first command and continue executing the next command until the last command is completed.
2. Set the number of executions (optional):

   If you choose the **Continuous Execution** method, you can check **Loop** and enter the number of loops as needed. After all commands are executed, it will continue to execute from the beginning until the loop count is completed.
3. Start Execution:
   - **Continuous Execution**: Click the **Start** button to execute the entire workflow.
   - **Step-by-Step Execution**: Click the **Start** button or double-click the command to execute the current step.
4. The timeline will display the current execution count and running time.

Button	Description
Start	Start executing the current workflow.
Pause	Pause the currently executing command.
Stop	Stop the currently executing command and exit execution state.

> **Note**
>
> After starting execution, you cannot switch execution modes, add, connect, or edit commands.

Debug Command

Skip: Temporarily ignore the execution of one or more commands during debugging, directly skipping these commands and continuing with subsequent commands.

Breakpoint: Set a breakpoint on a command during debugging; when execution reaches that command, the program will pause.

Inspection

With workflow orchestration, the software automates the entire process of scanning, inspecting, and reporting. This enables fully automated, batch processing of inspections, quickly providing accurate 3D data of the workpiece.

After finishing the scan and processing and saving the scan data in a grid format, add the inspection command, such as SHINING 3D Inspect. Configure all the paths and check **PDF** and **Open Report (PDF)**. When the process reaches this command and executes successfully, it will automatically save and display the inspection report in the **Inspection View**.

Workflow Example

Inspection Report Example

Command List

Turntable Command

Turntable Homing Routine

- Angle: 0 ~ 360°.
- Speed: 1 ~ 100%.
- Acceleration/ms/(60r/min): 80 ~ 9999.

Turntable Routine

- Angle: 0 ~ 360°.
- Relative Angle: Default is on, rotates the specified angle based on the current position.
- Speed: 1 ~ 100%.

- Acceleration/ms/(60r/min): 80 ~ 9999.

Robot Commands

> **Note**
>
> - The point refers to the final coordinates where the robot stops moving.
> - When recording the point, ensure that the robot is in an enabled state.

Homing Routine

Get Current Position: Obtain the current position as origin.

Linear Motion

The robot's end effector moves along a straight line.

- Speed: 1 ~ 100%.
- Acceleration: 1 ~ 2500 mm/s².
- Linear Speed: 1 ~ 2000 mm/s.
- Auto Scan: Once enabled, there's no need to add the **Start Scan** command; object will be scanned automatically upon reaching the designated position.
- Teaching Interval: Points are recorded every 5 seconds by default, and can be set up to every 60 seconds.
- Robot Path Teaching:

  ##### Manual

  After checking **Auto**, click the **Start Teaching**. Then physically move the robot arm to the desired position, and hold still for the duration set in the point recording interval; the software will add the point into point list.

  ##### 3Dconnexion SpaceMouse

  When the 3D mouse is connected, it defaults to recording points with the 3D mouse.

  Left-click to save the point, right-click to exit recording.

Joint Motion

The robot's end effector moves along an irregular curve.

- Speed: 1 ~ 100%.

- Acceleration: 1 ~ 360°/s².
- Joint Speed: 1 ~ 180°/s, default 90°/s.
- Auto Scan: Once enabled, there's no need to add the **Start Scan** command; object will be scanned automatically upon reaching the designated position.
- Teaching Interval: Points are recorded every 5 seconds by default, and can be set up to every 60 seconds.
- Robot Path Teaching:

  #### Manual

  After checking **Auto**, click the **Start Teaching**. Then physically move the robot arm to the desired position, and hold still for the duration set in the point recording interval; the software will add the point into point list.

  #### 3Dconnexion SpaceMouse

  When the 3D mouse is connected, it defaults to recording points with the 3D mouse.

  Left-click to save the point, right-click to exit recording.

> **Note**
>
> The joint motion command in the calibration interface can also capture image automatically.

General Motion

Set the movement mode at each point. The default movement mode is joint motion, which can be modified to linear motion.

- Auto Scan: Once enabled, there's no need to add the **Start Scan** command; object will be scanned automatically upon reaching the designated position.
- Teaching Interval: Points are recorded every 5 seconds by default, and can be set up to every 60 seconds.
- Robot Path Teaching:

  #### Manual

  After checking **Auto**, click the **Start Teaching**. Then physically move the robot arm to the desired position, and hold still for the duration set in the point recording interval; the software will add the point into point list.

  #### 3Dconnexion SpaceMouse

  When the 3D mouse is connected, it defaults to recording points with the 3D mouse.

  Left-click to save the point, right-click to exit recording.

Calibration Commands

The calibration commands **Wide Mode Calibration**, **Precise Mode Calibration**, **Wide Mode Advanced Calibration**, and **Precise Mode Advanced Calibration** are calibration templates.

> ⚠ **Caution**
>
> The calibration templates contain multiple preset combinations of calibration commands. When modifying the commands from the template, changes will not be saved to the template and you must manually save the solution.

Calibrate

- Calibration Mode: Choose **Calibration** or **Advanced Calibration**.
- Calibration Range: Choose **Rapid** or **HD**.

Calibration Settings

- Exposure: The longer the exposure time, the higher the image brightness.
- Camera Gain: The higher the gain, the higher the image brightness, but it may also introduce noise. It is recommended to use in low light environment. Default is synchronized left and right camera.

Deploy Calibration Board

Pop up the calibration board inside the base.

Stow Calibration Board

Stow the popped-up calibration board back into the base.

Capture Image

Capture images for calibration calculations.

Previous Step

Return to the previous calibration image capture command.

Scanner Commands

New Project Group

- Range: Select **Large Range** or **Small Range**.
- Name: Project group name.

  ☐ Keep Backup:

  - If checked, a backup file will be generated each time you save.
  - If not checked, multiple executions of the command will only keep the last created project.

- Default path: Set the default path for the project group to save.

Create Project

Range: Choose **Rapid** or **HD**.

Scan Setting

- Resolution: Low, Medium, High (default); for projects with large scanning data and lower precision requirements, the resolution can be appropriately reduced.
- Align Mode

Align Mode	Description
Markers	It is suitable for objects with fewer features or symmetrical features, as well as objects requiring high scanning accuracy. It requires attaching markers on the surface of the object.
Global Markers	It is suitable for objects with fewer features or symmetrical features, as well as objects requiring high scanning accuracy. It requires attaching markers on the surface of the object, and you can import global markers file (.p3 / .dgm / .txt / .asc) or scanning global markers directly. After switching to **Point Cloud**, you can tick **Add global markers** to automatically add the recognized global markers during the scanning process. **Note** There can only be one global markers file in a project. **Caution** The scanning method cannot be switched to scan global markers after scanning the point cloud data. If you need to scan global markers, please use the global markers mode first.

> **Note**
>
> If markers have been placed but no data can be collected, please check the condition of the placed markers (there should be at least 4 markers within each scanning area, and at least 4 common markers between adjacent scanning areas) or switch the align mode.

- Marker Size: Check the marker sizes that need to be recognized and markers will be displayed in the camera window during the scanning.
- Monocular-Stereo Fusion: For scanning objects with deep holes or recesses, it is recommended that you enable the binocular scanning function, which will retain more scanning data, specifically all data scanned by the left and right cameras.

> **Note**
>
> - If this function is disabled, only the common data scanned by the left and right cameras will be retained.
> - This function can be set only while the device is connected before or after scanning.

> **Caution**
>
> If data layering occurs during scanning, it is recommended that you perform recalibrate before enabling this function.

- Scan Box: When enabled (default), the field of view of the scanner's head will be displayed in real-time as a 3D translucent cone during each stage of scanning, assisting in checking the integrity of the scanning data.

> **Note**
>
> In the Markers / Global Markers align mode, the recognized markers will be displayed in real-time within the field of view.

- Object: Choose different scanning object modes based on the material of the object, including **Normal** and **Black/Reflective**.

Camera Settings

Exposure Mode: Set as automatic exposure by default. You can select among single exposure, double exposure, triple exposure, and custom exposure.

If automatic exposure is used, the center of the camera window will display exposure area, please drag it to the center of the object to be scanned to ensure exposure effect.

Point Cloud Exposure: If you need to adjust the brightness of the scanned point cloud, you can use the stepper to adjust the point cloud exposure level; the higher the exposure level, the longer the exposure time and the higher the brightness.

Marker Exposure: If you need to adjust the brightness of the scanned markers, you can adjust the marker exposure level; the higher the exposure level, the longer the exposure time and the higher the brightness, resulting in more recognized markers (marker recognition function must be enabled).

Camera Gain: For objects of different colors and materials, you can adjust the camera gain to obtain better scanning data; the gain is synchronized for left and right cameras. To set different gains, click .

Light Engine Brightness: If the object's material is highly reflective, it is recommended to lower the light engine brightness, i.e., the brightness of the stripes projected by the scanner.

Background Masking: If the scanning background is cluttered, you can enable background masking to automatically mask the background of the object to be scanned; The larger the value set, the larger the masking range.

Laser: When enabled (default), you can see laser points on the object to be scanned, and by checking whether the two projected laser points overlap, you can determine if the scanning distance is appropriate.

Real-time Marker Recognition: When enabled, you can view recognized markers in the camera window.

> **Note**
>
> - Previously scanned markers / markers recognized by the left and right cameras in the first frame will be displayed as green plus signs;
> - Newly recognized markers by the left and right cameras in this session (starting from the second frame) will be displayed as red plus signs;
> - Marker recognized by only one side camera or with deviation will not display numbers.
> - You can enhance recognition effects by adjusting exposure to recognize more markers.

Optical Focusing

If the projection is blurry or the scanning effect is not clear, you can use the optical machine to focus; before focusing, turn on the laser projector, move the object to the appropriate position, and ensure that the two laser points overlap.

After focusing the optical machine, recalibration is required.

Start Scan

Start scanning. If the robot command has enabled **Auto Scan**, this command can be omitted before the robot command.

End Scan

End scan. This command must be added after the command that completes the scan.

Projects Alignment

Add the **New Projects Group** command. Once the data scan is complete, add the **New Project** command. After the data scan is completed again, this command can be added to merge the scanned data from the two projects.

When this command is run, the scanning software will automatically open and enter the alignment interface. After the alignment is complete, you need to manually click **Restore** to continue executing the workflow.

Mesh Processing

This command is usually added after the **end scan** command. The **save data** command can only be successfully executed after the scan data has been meshed.

Mesh Parameters:

Icon	Function	Description
	Unwatertight	Unclosed model stays the way it is scanned, and its processing time is shorter than that of **Watertight**.
	Half watertight	Some of the holes will be filled automatically.
	Watertight	All holes will be filled automatically. The data can directly be 3D printed. The **Model Quality** can be set as **High**, **Med** (default), or **Low** only for the watertight mesh.

Mesh Processing:

- **Filter**: Optimize the data and improve the clarity of the data. Setting the filter level high will lose some small details.
  - **None**: No optimization.

  - **Standard**: Optimizes data slightly and preserves data characteristics.
  - **Med**: Reduce the noise on the surface of the scan data.
  - **High**: Reduce the noise on the surface of the scan data and sharpen it powerfully.
- **Smooth**: Smooth the possible noise on the surface of the scan data. Three optimization options are available: **Standard** (default), **Med** and **High**.
- **Remove Isolated Surface**: Remove small floating parts isolated from the main data. Adjust the area of floating parts to be removed by dragging the slider or using the stepper. The default value is 10, and the range is 0 ~ 1000 (0 indicates no removal). **For a specific illustration of the effect, refer to the following images:**
- **Maximum Number of Patches**: Set the max plate number to get mesh model's triangle plate number which is within configured plate number (enabled by default). Please input the value reasonably, avoiding entering too small values, as excessive simplification may result in lower data quality.
- **Automatic Hole Filling**: Auto fill the small holes when generating the mesh. The default value is 10 with a range of 0 to 100. You can delete unwanted neighboring areas to delete edge noise data; the default value is 3 with a range of 0 ~ 10 (0 indicates no neighboring area deleted). This function is unavailable for watertight models.
- **Remove Spike**: Remove spike-like data on the image edge.
- **Marker Hole Filling**: Fill in the surface of the object that is not scanned to the pasting markers. For the align mode of **Markers**, this function is enabled by default for watertight models; for other align modes, this function is unavailable.
- **Recommended Parameters**: When enabled, it will automatically use the recommended parameters for meshing.

Clear Data

Delete all scan data in the current project.

Save Data

Save Types: Supports STL, P3, ASC, OBJ, PLY, 3MF, and DGM formats.

- Name: The name of the model file.

☐ Keep Backup:

  - If checked, a backup file will be generated each time you save.
  - If not checked, multiple executions of the command will only keep the last created project.

☐ Use SN:

  After connecting to the external controller, the product SN will be transmitted to the RC software via the Modbus protocol, and the SN will be appended to the custom name when saving the STL file.

- Save Path: Click **Select** to choose a folder in the file explorer as the save path for the model file.

Inspect Commands

SHINING3D Inspect

- exe Program Path: Select the path where the SHINING3D Inspect exe program is located.
- Workspace path: Select the project file for SHINING3D Inspect (*.SNIProj), that is, inspection template.
- Report Export Path: Select the path to save the inspection report.
- Report Export Format

  - [x] PDF
    - [x] Open Report (PDF): Automatically open the inspection report in the Inspection interface after scanning is complete.
  - [ ] CSV

- [ ] Save inspection project
- [ ] Share Memory

PolyWorks

- Inspection Software: The default inspection software is PolyWorks.
- exe Program Path: Select the path where the PolyWorks exe program is located.
- Workspace Path: Select the path for the PolyWorks workspace.
- Report Export Path: Select the path to save the inspection report.
- Macro script: Select the path where the PolyWorks macro script is located.
- [x] Open inspection report

MES System Interface

Call external programs with parameters.

- External exe path: Select the path where the MES client exe file is located.
- External exe custom parameters: Add startup parameters. Special characters are not supported.

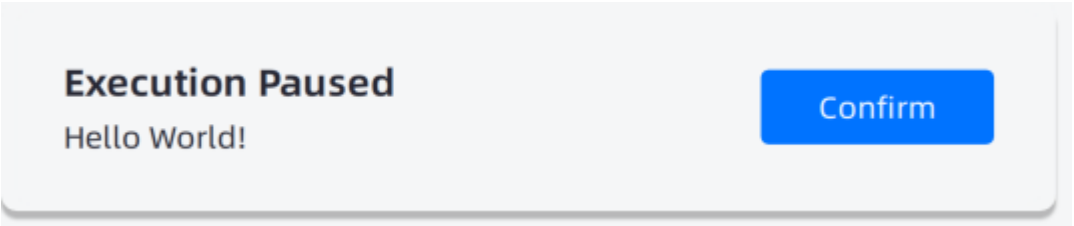
Other Commands

Waiting Time

Insert the specified wait time (ms) between commands.

Custom Prompt

Input text of 1 to 30 characters. When this command is executed, a text prompt will pop up, and the execution process will pause. Click confirm to continue the execution process.



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